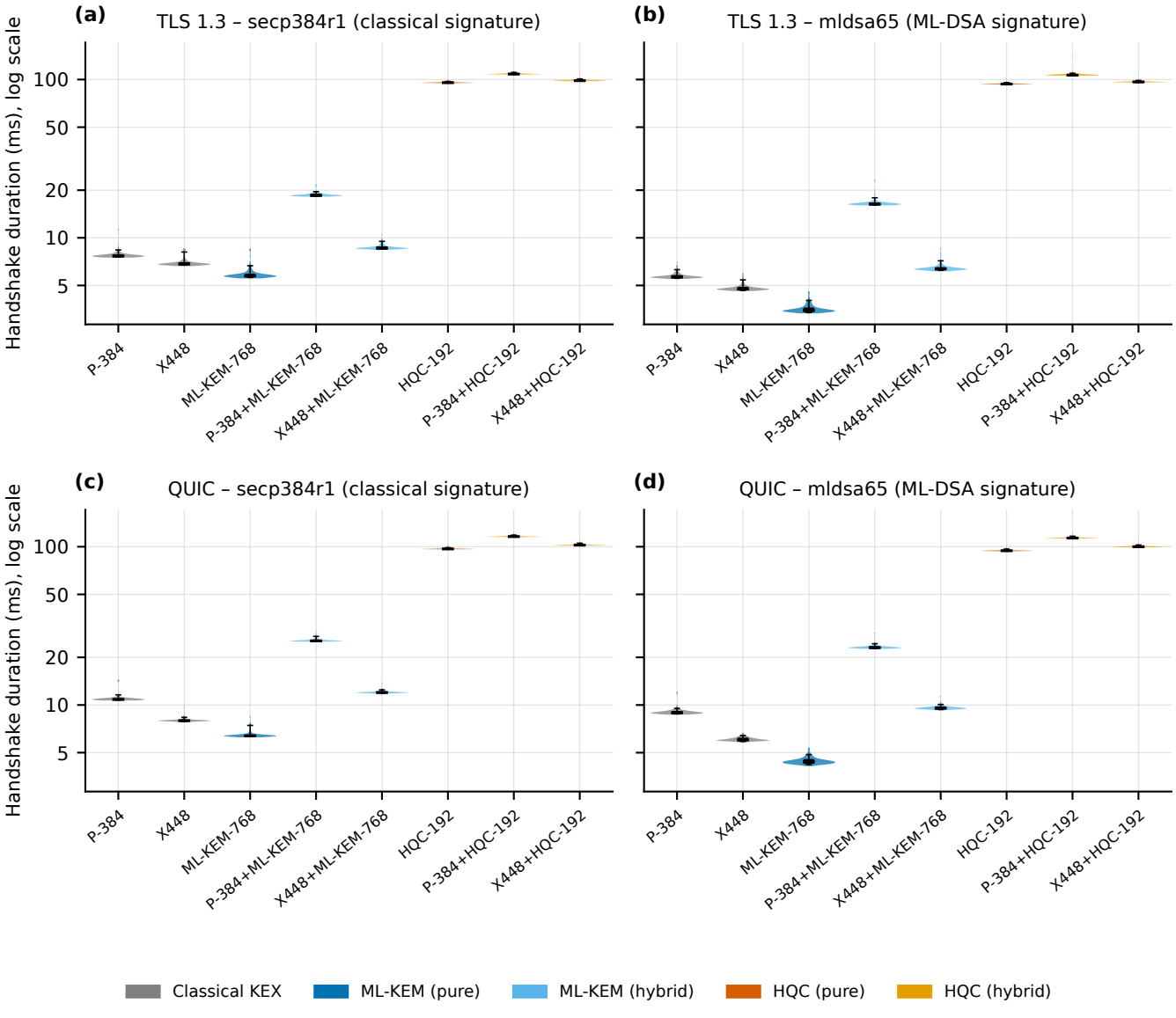

(a)
TLS 1.3 – secp384r1 (classical signature)
(b)
TLS 1.3 – mldsa65 (ML-DSA signature)
(c)
QUIC – secp384r1 (classical signature)
(d)
QUIC – mldsa65 (ML-DSA signature)
Handshake duration (ms), log scale
100
50
20
10
5
P-384
X448
ML-KEM-768
P-384+ML-KEM-768
X448+ML-KEM-768
HQC-192
P-384+HQC-192
X448+HQC-192
Classical KEX
ML-KEM (pure)
ML-KEM (hybrid)
HQC (pure)
HQC (hybrid)